

M000 Week 0 — English Worksheet (Beginner)

Topic: Pixel Art — Counting Squares · **Course:** Builder Basics · **Level:** Beginner · **Time:** about 30 minutes

A block in Minecraft has three numbers: **X** (left ↔ right), **Y** (up ↑ down), and **Z** (depth). On a flat wall, **Z stays the same** — you only change **X** and **Y**.

 This is a warm-up to do **before Week 1**. It helps you count squares.

1 · Read the Wall

Open the **Pixel Art** world. A flat wall has red pixel art on it.

- **X** = the columns, left to right →
- **Y** = the rows, bottom to top ↑
- **Z** = the depth. It stays the **same**, so we ignore it here.

The big idea:

 To color **one** square, the **start** and the **end** are the **same** square.

Find the smallest red mark on the wall. Circle one — is it 1 square or 2 squares? 1 · 2

2 · One Square: Start = End

X runs across the top. **Y** runs up the side. One square is red.

Y \ X	0	1	2	3	4
2					
1					
0					

The red square is at **X = 2, Y = 1**. Here is the command two ways:

Blocks

```
fill [  red wool ]  
  from world X 2 Y 1 Z 0  
  to world X 2 Y 1 Z 0
```

Python

```
blocks.fill(RED_WOOL, world(2, 1, 0), world(2, 1, 0), FillOperation.Replace)
```

The **from** and the **to** are the **same**.

In the python line, the from is **(2, 1, 0)**. What is the to?

3 · Count the Squares 1234

This line is flat. It sits on **Y = 1** and goes from **X = 1** to **X = 3**.

Y \ X	0	1	2	3	4
2					
1					
0					

Count the red squares. Circle one: 2 · 3 · 4

The line covers X = 1, then which numbers? Fill the gaps: 1 , __ , __

4 · Find the Difference 🐉

Both lines look almost the same. One colors **one** square. One colors **four**.

```
# Line A
blocks.fill(RED_WOOL, world(2, 2, 0), world(2, 2, 0), FillOperation.Replace)
```

```
# Line B
blocks.fill(RED_WOOL, world(2, 2, 0), world(3, 3, 0), FillOperation.Replace)
```

Which line colors only **ONE** square — A or B? Why?

5 · Fill the Gap

You want a flat line **3** squares long. It starts at **X = 1** and sits on **Y = 2**. One number is missing.

Y \ X	0	1	2	3	4
2					
1					

```
blocks.fill(RED_WOOL, world(1, 2, 0), world(__, 2, 0), FillOperation.Replace)
```

Word bank: 2 · 3 · 4

Write the missing number (look at the grid and count):

6 · Tell Me What You Built 📷

Now switch to your homework world. Color **one single square**, then make **one short line**. When you finish, come back here.

Send a photo or video, then explain what you did. Use these sentence starters — write 2 or 3 sentences.

First, I colored a square at $X \dots, Y \dots$

For one square, my start and end were ...

Then I made a line that was ... squares long.

7 · Record Your Walkthrough

Take a video on your phone (or a parent's phone) and show your wall. Talk like you are teaching a friend. Try to use these words: **X, Y, same, fill**.

1. Show the square you colored.
2. Say why the start and the end are the same.
3. Show your line and count the squares out loud.

Write what you will say in your video. You can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.